

Academic Regulations

Course Structure & Detailed Syllabus

CHOICE BASED CREDIT SYSTEM

R25

COMPUTER SCIENCE & ENGINEERING

(Data Science)

Bachelor of Technology (B.Tech)

For batches admitted from the academic year 2025-2026

**For batches admitted Lateral Entry Scheme from the
academic year 2026-2027**



**MARRI
LAXMAN
REDDY**

GROUP OF INSTITUTIONS

MLR Institute of Technology

(Autonomous)

**Laxman Reddy Avenue, Dundigal
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R25 COURSE STRUCTURE

For batches admitted from the academic year 2025-2026

**For batches admitted Lateral Entry Scheme from the
academic year 2026-2027**

MLR INSTITUTE OF TECHNOLOGY
COMPUTER SCIENCE & ENGINEERING -DATA SCIENCE
COURSE STRUCTURE – R25

I YEAR I SEMESTER									
Code	Cours e	Category	Periods per Week			Credits	Scheme of Examination MaximumMarks		
			L	T	P		Interna l	Externa l	Total
A7BS01	Matrices and Calculus	BSC	3	1	0	4	40	60	100
A7BS08	Advanced Engineering Physics	BSC	3	0	0	3	40	60	100
A7CS01	Programming for Problem Solving	ESC	3	0	0	3	40	60	100
A7EE04	Basic Electrical Engineering	ESC	3	0	0	3	40	60	100
A7ME02	Computer Aided Engineering Drawing	ESC	2	0	2	3	40	60	100
A7BS09	Advanced Engineering Physics Lab	BSC	0	0	2	1	40	60	100
A7CS02	Programming for Problem Solving Lab	ESC	0	0	2	1	40	60	100
A7EE05	Basic Electrical Engineering Lab	ESC	0	0	2	1	40	60	100
A7CS06	IOT and IT Workshop	ESC	0	0	2	1	40	60	100
	Induction Program								
TOTAL			14	1	10	20	360	540	900
I YEAR II SEMESTER									
Code	Course	Category	Periods per Week			Credits	Scheme of Examination MaximumMarks		
			L	T	P		Internal	External	Total
A7BS02	Ordinary Differential Equations and Vector Calculus	BSC	3	0	0	3	40	60	100
A7BS10	Engineering Chemistry	BSC	3	0	0	3	40	60	100
A7CS03	Data Structures	ESC	3	0	0	3	40	60	100
A7EC01	Electronic Devices and Applications	ESC	3	0	0	3	40	60	100
A7HS01	English for Skill Enhancement	HSMC	3	0	0	3	40	60	100
A7BS11	Engineering Chemistry Lab	BSC	0	0	2	1	40	60	100
A7CS04	Data Structures Lab	ESC	0	0	2	1	40	60	100
A7HS02	English Language and Communication Skills Lab	HSMC	0	0	2	1	40	60	100
A7ME03	Engineering Workshop	ESC	0	0	2	1	40	60	100

A7CS05	Python Programming Lab	ESC	0	0	2	1	40	60	100
TOTAL			15	0	10	20	400	600	1000

II YEAR I SEMESTER									
Code	Course	Category	Periods per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		CIE	SEE	Total
A7BS03	Mathematical and Statistical Foundations	BSC	3	0	0	3	40	60	100
A7CS08	Computer Organization and Architecture	PCC	3	0	0	3	40	60	100
A7CS09	Object Oriented Programming through java	PCC	3	0	0	3	40	60	100
A7CS11	Software Engineering	PCC	3	0	0	3	40	60	100
A7CS13	Data Base Management System	PCC	3	0	0	3	40	60	100
A7BS07	Computational Mathematics Lab	BSC	0	0	2	1	40	60	100
A7CS10	Object Oriented Programming through java Lab	PCC	0	0	2	1	40	60	100
A7CS12	Software Engineering Lab	PCC	0	0	2	1	40	60	100
A7CS14	Data Base Management System Lab	PCC	0	0	2	1	40	60	100
A7CS53	Skill Development(Web Technologies)	PCC	0	0	2	1	0	100	100
TOTAL			15	0	10	20	360	640	1000
II YEAR II SEMESTER									
Code	Course	Category	Periods per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		CIE	SEE	Total
A7CS07	Discrete Mathematics	PCC	3	0	0	3	40	60	100
A7CS15	Operating Systems	PCC	3	0	0	3	40	60	100
A7CS17	Algorithms Design and Analysis	PCC	3	0	0	3	40	60	100
A7CS18	Computer Networks	PCC	3	0	0	3	40	60	100
A7DS01	Introduction to Data Science	PCC	3	0	0	3	40	60	100
A7HS08	Innovation and Entrepreneurship	HSMC	2	0	0	2	40	60	100
A7CS16	Operating Systems lab	PCC	0	0	2	1	40	60	100
A7CS19	Computer Networks Lab	PCC	0	0	2	1	40	60	100
A7DS02	Data Science Lab	PCC	0	0	2	1	40	60	100
A7DS03	Skill Development(Data Visualization -R Programming/Power BI)	PCC	0	0	2	1	0	100	100
A7HS05	Indian Knowledge System	MC	1	0	0	1	50	0	50
TOTAL			18	0	8	22	410	640	1050

III YEAR I SEMESTER									
Code	Course	Category	Periods per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		CIE	SEE	Total
A7DS04	Predictive Analytics	PCC	3	0	0	3	40	60	100
A7DS22	Data Mining	PCC	3	0	0	3	40	60	100
A7AI03	Machine Learning	PCC	3	0	0	3	40	60	100
PEC-1	Professional Elective-I	PEC	3	0	0	3	40	60	100
OEC-1	Open Elective-I	OEC	2	0	0	2	40	60	100
A7DS05	Predictive Analytics Lab	PCC	0	0	2	1	40	60	100
A7DS23	Data Mining Lab	PCC	0	0	2	1	40	60	100
A7AI05	Machine Learning Lab	PCC	0	0	2	1	40	60	100
A7DS06	Field Based Research Project	PCC	0	0	4	2	0	100	100
A7CS54	Skill Development(UI Design – Flutter /Android Studio)	PCC	0	0	2	1	0	100	100
A7HS04	Gender Sensitization Lab*/ Human Values and Professional Ethics*	MC	1	0	0	1	50	0	50
TOTAL			15	0	12	21	370	680	1050
III YEAR II SEMESTER									
Code	Course	Category	Periods per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		CIE	SEE	Total
A7DS07	Data Stream Processing	PCC	3	0	0	3	40	60	100
A7DS08	Big data Technologies	PCC	3	0	0	3	40	60	100
A7HS06	Business Economics and Financial Analysis	HSM C	3	0	0	3	40	60	100
PE-11	Professional Elective-II	PEC	3	0	0	3	40	60	100
OE-II	Open Elective – II	OEC	2	0	0	2	40	60	100
A7DS09	Data Stream Processing Lab	PCC	0	0	2	1	40	60	100
A7DS10	Big data Technologies Lab	PCC	0	0	2	1	40	60	100
A7DS11	Exploratory Data Analysis Lab	PCC	0	0	2	1	40	60	100
A7HS03	English For Employability Skills Lab	HSC	0	0	2	1	40	60	100
A7AI14	Prompt Engineering	PCC	0	0	2	1			

							0	100	100
A7BS12	Environmental Science	MC	1	0	0	1	50	0	50
TOTAL			14	0	10	20	410	640	1050
*Note: For the courses Gender Sensitization Lab and Human Values and Professional Ethics - one hour of instruction will be conducted on alternate weeks. For example, if a one-hour class for Gender Sensitization Lab is conducted this week, then a one-hour class for Human Values and Professional Ethics will be conducted in the following week.									

IV YEAR I SEMESTER									
Code	Course	Category	Periods per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		CIE	SEE	Total
A7DS12	Data Stream Mining	PCC	3	0	0	3	40	60	100
A7DS13	Web and Social Media Analytics	PCC	3	0	0	3	40	60	100
A7HS07	Fundamentals of Management	HSM C	3	0	0	3	40	60	100
PEC-III	Professional Elective-III	PEC	3	0	0	3	40	60	100
PEC-IV	Professional Elective – IV	PEC	3	0	0	3	40	60	100
OEC-IV	Open Elective – III	OEC	2	0	0	2	40	60	100
A7DS14	Data Stream Mining Lab	PCC	0	0	2	1	40	60	100
A7DS15	Web and social media Analytics Lab	PCC	0	0	2	1	40	60	100
A7DS16	Industry Oriented Mini Project/ Summer Internship	PCC	0	0	4	2	0	100	100
TOTAL			17	0	08	21	320	580	900
IV YEAR II SEMESTER									
Code	Course	Category	Periods per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		CIE	SEE	Total
PE-V	Professional Elective – V	HSMC	3	0	0	3	40	60	100
PE-VI	Professional Elective – VI	PEC	3	0	0	3	40	60	100
A7DS17	Project Work	PCC	0	0	28	14	40	60	100
TOTAL			6	0	28	20	120	180	300

PROFESSIONAL ELECTIVES

PROFESSIONAL ELECTIVE – I			PROFESSIONAL ELECTIVE – II		
S. No	Code	Subject	S. No	Code	Subject
1	A7CS24	Computer Graphics	1	A7CS58	Image Processing
2	A7CS49	Graph Theory	2	A7CS30	Block chain Technology
3	A7CS25	Software Testing Methodologies	3	A7CS31	Software Project Management
4	A7CS50	Adhoc & Sensor Networks	4	A7DS18	Mining Massive Datasets
5	A7CS26	Web programming	5	A7CS32	Full Stack Development
6	A7CS27	Distributed Systems	6	A7AI09	Generative AI
PROFESSIONAL ELECTIVE – III			PROFESSIONAL ELECTIVE - IV		
S. No	Code	Subject	S. No	Code	Subject
1	A7AI20	Computer Vision	1	A7AI21	Augmented Reality & Virtual Reality
2	A7CS38	Vulnerability Assessment and Penetration Testing	2	A7CS41	Agile Methodology
3	A7CS22	DevOps	3	A7DS19	Time Series Analysis and Forecasting
4	A7CS28	Cryptography and Network Security	4	A7AI22	Quantum Computing
5	A7CS39	Cloud Computing	5	A7AI23	Robotic Process Automation
6	A7CS40	Information Retrieval Systems	6	A7CS42	Cyber Forensics
PROFESSIONAL ELECTIVE – V			PROFESSIONAL ELECTIVE - VI		
S. No	Code	Subject	S. No	Code	Subject
1	A7DS20	Social Media Mining	1	A7CS47	High Performance Computing
2	A7CS44	Nature Inspired Computing	2	A7CS48	Edge Computing
3	A7AI24	Ethics of Artificial Intelligence	3	A7CS37	Scripting Languages
4	A7AI25	Game Theory	4	A7DS24	UI/UX
5	A7CS45	Mobile Application Development	5	A7CS51	Sustainable Engineering
6	A7CS46	Human Computer Interaction	6	A7CS52	Distributed Databases

OPEN ELECTIVES**OPEN ELECTIVE-I**

S. No	Code	Subject
1	A7DS25	Fundamentals of AI
2	A7DS21	R Programming

OPEN ELECTIVE-II

S. No	Code	Subject
1	A7DS26	Introduction to Natural Language Processing
2	A7DS27	MERN Stack Development

OPEN ELECTIVE-III

S. No	Code	Subject
1	A7DS28	Chatbots
2	A7DS29	Android Application Development